MATHEMATICS
ANNOTATED READING NOTES

PDE

Partial Differential Equations

英文精读札记与中文边注
目录骨架 / Outline Edition

Reading companion to Lawrence C. Evans
Partial Differential Equations, Second Edition

基于 Evans 原著的学习整理

$$\partial_t u - \Delta u = 0$$

LUMENBOOK

2026-11-09

澄明讲义 / 数学阅读札记

How to Use These Notes

These notes follow Lawrence C. Evans's *Partial Differential Equations*, second edition. The English PDF supplies the chapter order and mathematical source. The supplied Chinese translation is a reference for terminology and reading support.

English text and margin notes

Read the main argument first. Margin notes will explain selected vocabulary, long sentences, and mathematical ideas. They will be short enough to remain beside the relevant passage. Longer derivations belong in the main text.

Expanded proofs

An expanded proof will state the assumptions, explain the strategy, and fill in omitted calculations. Each use of approximation, integration by parts, compactness, or passage to a limit must identify the applicable hypotheses. Source theorem numbers will be recorded explicitly; the notes' own numbering does not automatically coincide with the original book.

Current status

This is an outline edition. Chapter and section files exist, but the book's English exposition, proofs, and exercises have not yet been transcribed or expanded. The margin notes on this page demonstrate the annotation system.

The introduction precedes Part I, as in the source. The three parts contain Chapters 2–4, 5–7, and 8–12. Appendices A–E supply background material. The bilingual glossary is an added study aid. Detailed subsection headings will be added when their source pages are studied.

中文提示

正文以英文为主，必要时用中文边注解释词句和数学难点。中译本辅助理解；遇到版本、符号或条件差异，须回到英文原文核对。

Language / 词句

provided that

意为“只要”或“在……条件下”。阅读定理时，应把这类表达后面的内容识别为假设。

Idea / 难点

边注适合交代“这一步为什么成立”或提示先修知识。需要多行推导的证明补充放在正文，避免边栏过密。

Contents

How to Use These Notes ii

Preface to the second edition viii

Preface to the first edition ix

1 Introduction 1

§ 1.1 Partial Differential Equations 1

§ 1.2 Examples 1

§ 1.3 Strategies for Studying PDE 1

§ 1.4 Overview 1

§ 1.5 Problems 1

§ 1.6 References 1

I Representation Formulas for Solutions 2

2 Four Important Linear Partial Differential Equations 3

§ 2.1 Transport Equation 3

§ 2.2 Laplace's Equation 3

§ 2.3 Heat Equation 3

§ 2.4 Wave Equation 3

§ 2.5 Problems 3

§ 2.6 References 4

3 Nonlinear First-Order PDE 5

§ 3.1 Complete Integrals, Envelopes 5

§ 3.2 Characteristics 5

§ 3.3 Introduction to Hamilton–Jacobi Equations 5

§ 3.4 Introduction to Conservation Laws 5

§ 3.5 Problems 5

§ 3.6 References 5

4 Other Ways to Represent Solutions 6

§ 4.1 Separation of Variables . . . 6
§ 4.2 Similarity Solutions . . . 6
§ 4.3 Transform Methods . . . 6
§ 4.4 Converting Nonlinear into Linear PDE . . . 6
§ 4.5 Asymptotics . . . 6
§ 4.6 Power Series . . . 6
§ 4.7 Problems . . . 7
§ 4.8 References . . . 7

II Theory for Linear Partial Differential Equations 8

5 Sobolev Spaces 9

§ 5.1 Hölder Spaces . . . 9
§ 5.2 Sobolev Spaces . . . 9
§ 5.3 Approximation . . . 9
§ 5.4 Extensions . . . 9
§ 5.5 Traces . . . 9
§ 5.6 Sobolev Inequalities . . . 9
§ 5.7 Compactness . . . 10
§ 5.8 Additional Topics . . . 10
§ 5.9 Other Spaces of Functions . . . 10
§ 5.10 Problems . . . 10
§ 5.11 References . . . 10

6 Second-Order Elliptic Equations 11

§ 6.1 Definitions . . . 11
§ 6.2 Existence of Weak Solutions . . . 11
§ 6.3 Regularity . . . 11
§ 6.4 Maximum Principles . . . 11
§ 6.5 Eigenvalues and Eigenfunctions . . . 11
§ 6.6 Problems . . . 11
§ 6.7 References . . . 12

7 Linear Evolution Equations 13

§ 7.1 Second-Order Parabolic Equations . . . 13
§ 7.2 Second-Order Hyperbolic Equations . . . 13

§ 7.3 Systems of Hyperbolic First-Order Equations 13
§ 7.4 Semigroup Theory 13
§ 7.5 Problems 13
§ 7.6 References 13

III Theory for Nonlinear Partial Differential Equations 14

8 The Calculus of Variations 15
§ 8.1 Introduction 15
§ 8.2 Existence of Minimizers 15
§ 8.3 Regularity 15
§ 8.4 Constraints 15
§ 8.5 Critical Points 15
§ 8.6 Invariance, Noether's Theorem 15
§ 8.7 Problems 16
§ 8.8 References 16

9 Nonvariational Techniques 17
§ 9.1 Monotonicity Methods 17
§ 9.2 Fixed Point Methods 17
§ 9.3 Method of Subsolutions and Supersolutions 17
§ 9.4 Nonexistence of Solutions 17
§ 9.5 Geometric Properties of Solutions 17
§ 9.6 Gradient Flows 17
§ 9.7 Problems 18
§ 9.8 References 18

10 Hamilton–Jacobi Equations 19
§ 10.1 Introduction, Viscosity Solutions 19
§ 10.2 Uniqueness 19
§ 10.3 Control Theory, Dynamic Programming 19
§ 10.4 Problems 19
§ 10.5 References 19

11 Systems of Conservation Laws 20
§ 11.1 Introduction 20
§ 11.2 Riemann's Problem 20

§ 11.3 Systems of Two Conservation Laws . . . 20
§ 11.4 Entropy Criteria . . . 20
§ 11.5 Problems . . . 20
§ 11.6 References . . . 20

12 Nonlinear Wave Equations 21

§ 12.1 Introduction . . . 21
§ 12.2 Existence of Solutions . . . 21
§ 12.3 Semilinear Wave Equations . . . 21
§ 12.4 Critical Power Nonlinearity . . . 21
§ 12.5 Nonexistence of Solutions . . . 21
§ 12.6 Problems . . . 21
§ 12.7 References . . . 22

Appendices 24

A Notation 24

§ A.1 Notation for matrices . . . 24
§ A.2 Geometric notation . . . 24
§ A.3 Notation for functions . . . 24
§ A.4 Vector-valued functions . . . 24
§ A.5 Notation for estimates . . . 24
§ A.6 Some comments about notation . . . 24

B Inequalities 25

§ B.1 Convex functions . . . 25
§ B.2 Useful inequalities . . . 25

C Calculus 26

§ C.1 Boundaries . . . 26
§ C.2 Gauss–Green Theorem . . . 26
§ C.3 Polar coordinates, coarea formula . . . 26
§ C.4 Moving regions . . . 26
§ C.5 Convolution and smoothing . . . 26
§ C.6 Inverse Function Theorem . . . 26
§ C.7 Implicit Function Theorem . . . 27
§ C.8 Uniform convergence . . . 27

D Functional Analysis **28**

§ D.1 Banach spaces . . . 28
§ D.2 Hilbert spaces . . . 28
§ D.3 Bounded linear operators . . . 28
§ D.4 Weak convergence . . . 28
§ D.5 Compact operators, Fredholm theory . . . 28
§ D.6 Symmetric operators . . . 28

E Measure Theory **29**

§ E.1 Lebesgue measure . . . 29
§ E.2 Measurable functions and integration . . . 29
§ E.3 Convergence theorems for integrals . . . 29
§ E.4 Differentiation . . . 29
§ E.5 Banach space-valued functions . . . 29

Bibliography **30**

Index **31**

Bilingual Glossary / 英汉术语表 **32**

Preface to the second edition

Reading source

Source preface

English printed p. xi; PDF page 12.

Outline only. Text, annotations, and proof details are pending.

Preface to the first edition

Reading source

Source preface

English printed p. xiii; PDF page 14.

Outline only. Text, annotations, and proof details are pending.

Chapter 1

Introduction

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 1; printed p. 1; PDF page 18.

Chinese reference: 中译本对应章：引言，书内第 1 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

1.1 Partial Differential Equations

Outline only. Text, annotations, and proof details are pending.

1.2 Examples

Outline only. Text, annotations, and proof details are pending.

1.3 Strategies for Studying PDE

Outline only. Text, annotations, and proof details are pending.

1.4 Overview

Outline only. Text, annotations, and proof details are pending.

1.5 Problems

Outline only. Text, annotations, and proof details are pending.

1.6 References

Outline only. Text, annotations, and proof details are pending.

Part I

Representation Formulas for Solutions

Chapter 2

Four Important Linear Partial Differential Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 2; printed p. 17; PDF page 34.

Chinese reference: 中译本对应章：四个重要的线性偏微分方程，书内第 13 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

2.1 Transport Equation

Outline only. Text, annotations, and proof details are pending.

2.2 Laplace's Equation

Outline only. Text, annotations, and proof details are pending.

2.3 Heat Equation

Outline only. Text, annotations, and proof details are pending.

2.4 Wave Equation

Outline only. Text, annotations, and proof details are pending.

2.5 Problems

Outline only. Text, annotations, and proof details are pending.

2.6 References

Outline only. Text, annotations, and proof details are pending.

Chapter 3

Nonlinear First-Order PDE

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 3; printed p. 89; PDF page 106.

Chinese reference: 中译本对应章：非线性一阶偏微分方程，书内第 61 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

3.1 Complete Integrals, Envelopes

Outline only. Text, annotations, and proof details are pending.

3.2 Characteristics

Outline only. Text, annotations, and proof details are pending.

3.3 Introduction to Hamilton–Jacobi Equations

Outline only. Text, annotations, and proof details are pending.

3.4 Introduction to Conservation Laws

Outline only. Text, annotations, and proof details are pending.

3.5 Problems

Outline only. Text, annotations, and proof details are pending.

3.6 References

Outline only. Text, annotations, and proof details are pending.

Chapter 4

Other Ways to Represent Solutions

> Reading source
> **Evans, *Partial Differential Equations*, second edition**
> English source: Chapter 4; printed p. 159; PDF page 176.
> Chinese reference: 中译本对应章：解的其他表示方法，书内第 111 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

4.1 Separation of Variables

Outline only. Text, annotations, and proof details are pending.

4.2 Similarity Solutions

Outline only. Text, annotations, and proof details are pending.

4.3 Transform Methods

Outline only. Text, annotations, and proof details are pending.

4.4 Converting Nonlinear into Linear PDE

Outline only. Text, annotations, and proof details are pending.

4.5 Asymptotics

Outline only. Text, annotations, and proof details are pending.

4.6 Power Series

Outline only. Text, annotations, and proof details are pending.

4.7 Problems

Outline only. Text, annotations, and proof details are pending.

4.8 References

Outline only. Text, annotations, and proof details are pending.

Part II

Theory for Linear Partial Differential Equations

Chapter 5

Sobolev Spaces

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 5; printed p. 239; PDF page 256.

Chinese reference: 中译本对应章：Sobolev 空间，书内第 169 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

5.1 Hölder Spaces

Outline only. Text, annotations, and proof details are pending.

5.2 Sobolev Spaces

Outline only. Text, annotations, and proof details are pending.

5.3 Approximation

Outline only. Text, annotations, and proof details are pending.

5.4 Extensions

Outline only. Text, annotations, and proof details are pending.

5.5 Traces

Outline only. Text, annotations, and proof details are pending.

5.6 Sobolev Inequalities

Outline only. Text, annotations, and proof details are pending.

5.7 Compactness

Outline only. Text, annotations, and proof details are pending.

5.8 Additional Topics

Outline only. Text, annotations, and proof details are pending.

5.9 Other Spaces of Functions

Outline only. Text, annotations, and proof details are pending.

5.10 Problems

Outline only. Text, annotations, and proof details are pending.

5.11 References

Outline only. Text, annotations, and proof details are pending.

Chapter 6

Second-Order Elliptic Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 6; printed p. 293; PDF page 310.

Chinese reference: 中译本对应章：二阶椭圆型方程，书内第 208 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

6.1 Definitions

Outline only. Text, annotations, and proof details are pending.

6.2 Existence of Weak Solutions

Outline only. Text, annotations, and proof details are pending.

6.3 Regularity

Outline only. Text, annotations, and proof details are pending.

6.4 Maximum Principles

Outline only. Text, annotations, and proof details are pending.

6.5 Eigenvalues and Eigenfunctions

Outline only. Text, annotations, and proof details are pending.

6.6 Problems

Outline only. Text, annotations, and proof details are pending.

6.7 References

Outline only. Text, annotations, and proof details are pending.

Chapter 7

Linear Evolution Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 7; printed p. 349; PDF page 366.

Chinese reference: 中译本对应章：线性发展方程，书内第 248 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

7.1 Second-Order Parabolic Equations

Outline only. Text, annotations, and proof details are pending.

7.2 Second-Order Hyperbolic Equations

Outline only. Text, annotations, and proof details are pending.

7.3 Systems of Hyperbolic First-Order Equations

Outline only. Text, annotations, and proof details are pending.

7.4 Semigroup Theory

Outline only. Text, annotations, and proof details are pending.

7.5 Problems

Outline only. Text, annotations, and proof details are pending.

7.6 References

Outline only. Text, annotations, and proof details are pending.

Part III

Theory for Nonlinear Partial Differential Equations

Chapter 8

The Calculus of Variations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 8; printed p. 427; PDF page 444.

Chinese reference: 中译本对应章：变分法，书内第 303 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

8.1 Introduction

Outline only. Text, annotations, and proof details are pending.

8.2 Existence of Minimizers

Outline only. Text, annotations, and proof details are pending.

8.3 Regularity

Outline only. Text, annotations, and proof details are pending.

8.4 Constraints

Outline only. Text, annotations, and proof details are pending.

8.5 Critical Points

Outline only. Text, annotations, and proof details are pending.

8.6 Invariance, Noether's Theorem

Outline only. Text, annotations, and proof details are pending.

8.7 Problems

Outline only. Text, annotations, and proof details are pending.

8.8 References

Outline only. Text, annotations, and proof details are pending.

Chapter 9

Nonvariational Techniques

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 9; printed p. 495; PDF page 512.

Chinese reference: 中译本对应章：非变分技术，书内第 350 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

9.1 Monotonicity Methods

Outline only. Text, annotations, and proof details are pending.

9.2 Fixed Point Methods

Outline only. Text, annotations, and proof details are pending.

9.3 Method of Subsolutions and Supersolutions

Outline only. Text, annotations, and proof details are pending.

9.4 Nonexistence of Solutions

Outline only. Text, annotations, and proof details are pending.

9.5 Geometric Properties of Solutions

Outline only. Text, annotations, and proof details are pending.

9.6 Gradient Flows

Outline only. Text, annotations, and proof details are pending.

9.7 Problems

Outline only. Text, annotations, and proof details are pending.

9.8 References

Outline only. Text, annotations, and proof details are pending.

Chapter 10

Hamilton–Jacobi Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 10; printed p. 543; PDF page 560.

Chinese reference: 中译本对应章：Hamilton–Jacobi 方程，书内第 384 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

10.1 Introduction, Viscosity Solutions

Outline only. Text, annotations, and proof details are pending.

10.2 Uniqueness

Outline only. Text, annotations, and proof details are pending.

10.3 Control Theory, Dynamic Programming

Outline only. Text, annotations, and proof details are pending.

10.4 Problems

Outline only. Text, annotations, and proof details are pending.

10.5 References

Outline only. Text, annotations, and proof details are pending.

Chapter 11

Systems of Conservation Laws

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 11; printed p. 571; PDF page 588.

Chinese reference: 中译本对应章：守恒律组，书内第 404 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

11.1 Introduction

Outline only. Text, annotations, and proof details are pending.

11.2 Riemann's Problem

Outline only. Text, annotations, and proof details are pending.

11.3 Systems of Two Conservation Laws

Outline only. Text, annotations, and proof details are pending.

11.4 Entropy Criteria

Outline only. Text, annotations, and proof details are pending.

11.5 Problems

Outline only. Text, annotations, and proof details are pending.

11.6 References

Outline only. Text, annotations, and proof details are pending.

Chapter 12

Nonlinear Wave Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 12; printed p. 617; PDF page 634.

Chinese reference: 中译本对应章：非线性波动方程，书内第 437 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

12.1 Introduction

Outline only. Text, annotations, and proof details are pending.

12.2 Existence of Solutions

Outline only. Text, annotations, and proof details are pending.

12.3 Semilinear Wave Equations

Outline only. Text, annotations, and proof details are pending.

12.4 Critical Power Nonlinearity

Outline only. Text, annotations, and proof details are pending.

12.5 Nonexistence of Solutions

Outline only. Text, annotations, and proof details are pending.

12.6 Problems

Outline only. Text, annotations, and proof details are pending.

12.7 References

Outline only. Text, annotations, and proof details are pending.

Appendices

Appendix

A

Notation

> Reading source
> **Evans, *Partial Differential Equations*, second edition**
> English source: Appendix A; printed p. 653; PDF page 670.
> Chinese reference: 中译本对应章：记号，书内第 463 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

A.1 Notation for matrices

Outline only. Text, annotations, and proof details are pending.

A.2 Geometric notation

Outline only. Text, annotations, and proof details are pending.

A.3 Notation for functions

Outline only. Text, annotations, and proof details are pending.

A.4 Vector-valued functions

Outline only. Text, annotations, and proof details are pending.

A.5 Notation for estimates

Outline only. Text, annotations, and proof details are pending.

A.6 Some comments about notation

Outline only. Text, annotations, and proof details are pending.

Appendix

B

Inequalities

> Reading source
>
> **Evans, *Partial Differential Equations*, second edition**
>
> English source: Appendix B; printed p. 661; PDF page 678.
>
> Chinese reference: 中译本对应章：不等式，书内第 471 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

B.1 Convex functions

Outline only. Text, annotations, and proof details are pending.

B.2 Useful inequalities

Outline only. Text, annotations, and proof details are pending.

Appendix

C

Calculus

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix C; printed p. 665; PDF page 682.

Chinese reference: 中译本对应章：微积分，书内第 475 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

C.1 Boundaries

Outline only. Text, annotations, and proof details are pending.

C.2 Gauss–Green Theorem

Outline only. Text, annotations, and proof details are pending.

C.3 Polar coordinates, coarea formula

Outline only. Text, annotations, and proof details are pending.

C.4 Moving regions

Outline only. Text, annotations, and proof details are pending.

C.5 Convolution and smoothing

Outline only. Text, annotations, and proof details are pending.

C.6 Inverse Function Theorem

Outline only. Text, annotations, and proof details are pending.

C.7 Implicit Function Theorem

Outline only. Text, annotations, and proof details are pending.

C.8 Uniform convergence

Outline only. Text, annotations, and proof details are pending.

Appendix

D

Functional Analysis

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix D; printed p. 674; PDF page 691.

Chinese reference: 中译本对应章：泛函分析，书内第 482 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

D.1 Banach spaces

Outline only. Text, annotations, and proof details are pending.

D.2 Hilbert spaces

Outline only. Text, annotations, and proof details are pending.

D.3 Bounded linear operators

Outline only. Text, annotations, and proof details are pending.

D.4 Weak convergence

Outline only. Text, annotations, and proof details are pending.

D.5 Compact operators, Fredholm theory

Outline only. Text, annotations, and proof details are pending.

D.6 Symmetric operators

Outline only. Text, annotations, and proof details are pending.

Appendix

E

Measure Theory

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix E; printed p. 684; PDF page 701.

Chinese reference: 中译本对应章：测度论，书内第 490 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

E.1 Lebesgue measure

Outline only. Text, annotations, and proof details are pending.

E.2 Measurable functions and integration

Outline only. Text, annotations, and proof details are pending.

E.3 Convergence theorems for integrals

Outline only. Text, annotations, and proof details are pending.

E.4 Differentiation

Outline only. Text, annotations, and proof details are pending.

E.5 Banach space-valued functions

Outline only. Text, annotations, and proof details are pending.

Bibliography

The source bibliography starts on English printed p. 689 (PDF page 706). Bibliographic entries and any additional references remain to be entered.

Outline only. Text, annotations, and proof details are pending.

Index

The source index starts on English printed p. 703 (PDF page 720). An index for these notes will be prepared as the text is written; page numbers must refer to this edition.

Outline only. Text, annotations, and proof details are pending.

Bilingual Glossary
英汉术语表

This added study aid will collect the English terms used in the notes, their Chinese equivalents, and links to the relevant definitions. Entries remain to be prepared.

Outline only. Text, annotations, and proof details are pending.